MPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EIE PART II: MEng, BEng and ACGI

Corrected Copy

COMPUTER ARCHITECTURE

Tuesday, 14 May 2:00 pm

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer TWO questions.

Answer ALL questions.

Use separate answer books for Sections A and B.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : S. Ben Ismail, S. Ben Ismail

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COMPUTER ARCHITECTURE 2

1. a) Describe the difference between a write-back and a write-through cache policy.

[2]

b) Give two reasons for having separate instruction and data caches in a CPU.

[2]

c) Consider a CPU attached to an n-way set-associative cache with 64 bytes/block and 16 blocks. A function executing on the CPU reads data at byte addresses $1024 + 24i$ for $i \in \{0, 1, 2, \ldots\}$.

i) What is the capacity of the cache in w-bit words?

[2]

ii) What type(s) of locality can be exploited by the cache during a single execution of the function? Only consider the memory accesses of the given data read sequence.

[2]

iii) Assuming that the cache is direct-mapped, estimate the combinatorial delay of a read access. State any extra assumptions that you need to make.

[3]

iv) Now assume that the cache is fully associative and starts cold (all blocks invalid). What is the first value of i that will cause an eviction?

[3]

v) Give an example of high-level C code involving both structs and arrays which might give rise to this address pattern.

[3]

vi) Describe how the C code could be optimised or restructured to improve locality, and estimate the resultant performance improvement.

[3]

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2. In this question, assume the use of the MIPS-I instruction set architecture.

a) Give two advantages of having a hard-wired zero register.

[3]

b) Give a sequence of assembly instructions which evaluates `$r1 += $r2*$r3`, where the three values are all unsigned 32-bit integers and are contained in registers with the same names.

[3]

c) Describe the instruction encoding of `ADDU` and `MULTU`. You should indicate the widths and meanings of the different instruction fields, but do not need to give the bit representations of constants.

[4]

d) In a particular embedded application, an engineer has found that the CPU spends most of its time calculating `x += y*C`, where `C` is a compile-time constant with at most 6 non-zero bits.

i) Propose a new custom instruction which could be used optimise this type of code, including its instruction encoding.

[3]

ii) Describe two different approaches for integrating your new instruction into the standard pipelined MIPS datapath, and give the performance advantages of each.

[4]

iii) Pick one approach from the previous answer, and describe any new hazards which could occur and whether each of those hazards should be handled using forwarding and/or stalls.

[3]